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Qualification Pack

AI - Machine learning Developer

QP Code: ASP/ITS/Q0101

Version: 2.0

NSQF Level: 5

Additional Skill Acquisition Programme Kerala || Additional Skill Acquisition Programme Kerala, 3rd
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Qualification Pack

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ASP/ITS/Q0101: AI - Machine learning Developer

Brief Job Description

Individuals in this role are responsible for designing, developing, and deploying AI and Machine Learning applications and platforms. They evaluate the performance of algorithmic models within the deployment environment and ensure optimal system integration. Their responsibilities include building, testing, and implementing end-to-end AI solutions tailored to specific use cases.

Personal Attributes

A learner in the AI and Machine Learning course should possess analytical and logical thinking, a strong problem-solving mindset, and curiosity for innovation. Attention to detail, patience, and perseverance are essential for handling complex data and algorithms. Mathematical aptitude, adaptability, and a commitment to continuous learning help in mastering evolving technologies.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [ASP/ITS/N0107: Python Programming with Scientific and Mathematical Applications](#)
2. [ASP/ITS/N0105: Core Concepts in Machine Learning](#)
3. [ASP/ITS/N0103: Bayesian methods And Text Mining](#)
4. [ASP/ITS/N0106: Deep Learning Foundations to Prompt Engineering](#)
5. [ASP/ITS/N0108: AI in the Cloud: Deployment, Ethics & Intelligent Agents.](#)
6. [ASP/ITS/N0109: On the Job Training for AI ML Developer.](#)
7. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Information Technology Sector
Sub-Sector	Artificial Intelligence
Occupation	Artificial Intelligence & Machine Learning, Artificial Intelligence & Machine Learning
Country	India

Qualification Pack

NSQF Level	5
Credits	18
Aligned to NCO/ISCO/ISIC Code	2512.0800
Minimum Educational Qualification & Experience	Diploma ((UG Diploma)CS/IT/EC and related Engineering branches) with NA of experience OR Completed 3 year diploma after 10th (CS/IT/EC and related Engineering branches) with 1.5 years of experience Data science, AI, ML, Data Analytics, related IT job roles OR 12th Class (or equivalent) with 3 Years of experience Data science, AI, ML, Data Analytics, related IT job roles OR Previous relevant Qualification of NSQF Level (4.5) with 1.5 years of experience Data science, AI, ML, Data Analytics, related IT job roles OR Previous relevant Qualification of NSQF Level (4) with 3 Years of experience Data science, AI, ML, Data Analytics, related IT job roles
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NIL
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	07/10/2028
NSQC Approval Date	07/10/2025
Version	2.0
Reference code on NQR	QG-05-IT-04446-2025-V2-ASAP
NQR Version	2

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ASP/ITS/N0107: Python Programming with Scientific and Mathematical Applications

Description

Learners will acquire foundational to intermediate proficiency in Python, enabling them to apply programming skills to scientific and mathematical problem-solving, data processing, and model development using essential libraries.

Scope

The scope covers the following :

- Learners will acquire foundational to intermediate proficiency in Python, enabling them to apply programming skills to scientific and mathematical problem-solving, data processing, and model development using essential libraries.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Write and debug Python programs using correct syntax, data types, control structures, and file operations.
- PC2.** Use NumPy and SciPy libraries to perform array manipulations and scientific computations.
- PC3.** Create and interpret visualizations using Matplotlib for scientific data.
- PC4.** Connect to databases using Python and perform data insertion, retrieval, and updates.
- PC5.** Solve linear algebra problems using Python for scientific and engineering applications.
- PC6.** Simulate and analyze random processes using appropriate Python libraries and techniques.
- PC7.** Formulate and solve basic optimization problems using tools like SciPy and PuLP.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	7	7	-	6
PC1. Write and debug Python programs using correct syntax, data types, control structures, and file operations.	1	1	-	1
PC2. Use NumPy and SciPy libraries to perform array manipulations and scientific computations.	1	1	-	1
PC3. Create and interpret visualizations using Matplotlib for scientific data.	1	1	-	-
PC4. Connect to databases using Python and perform data insertion, retrieval, and updates.	1	1	-	1
PC5. Solve linear algebra problems using Python for scientific and engineering applications.	1	1	-	1
PC6. Simulate and analyze random processes using appropriate Python libraries and techniques.	1	1	-	1
PC7. Formulate and solve basic optimization problems using tools like SciPy and PuLP.	1	1	-	1
NOS Total	7	7	-	6

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National Occupational Standards (NOS) Parameters

NOS Code	ASP/ITS/N0107
NOS Name	Python Programming with Scientific and Mathematical Applications
Sector	Information Technology Sector
Sub-Sector	
Occupation	Artificial Intelligence & Machine Learning
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

ASP/ITS/N0105: Core Concepts in Machine Learning

Description

Learners will gain an in-depth understanding of supervised and unsupervised machine learning algorithms, enabling them to develop, train, evaluate, and interpret predictive models on real-world datasets.

Scope

The scope covers the following :

- Building, training, and validating ML models on real-world datasets.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Explain key machine learning concepts, types of learning, and common algorithms.
- PC2.** Build and evaluate regression models using techniques like linear and polynomial regression on real-world data.
- PC3.** Apply classification algorithms such as logistic regression, decision trees, and SVM to labeled datasets.
- PC4.** Use non-parametric techniques like K-Nearest Neighbors for classification tasks with unknown data distributions.
- PC5.** Perform clustering and dimensionality reduction using unsupervised learning methods like k-means and PCA.
- PC6.** Evaluate model performance using metrics such as confusion matrix, cross-validation, precision-recall, and ROC curves.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Core concepts of supervised unsupervised and non parametric learning methods
- KU2.** Functioning and applications of algorithms like linear polynomial regression logistic regression decision trees SVM KNN k means and PCA
- KU3.** Principles of overfitting underfitting and bias variance tradeoff
- KU4.** Model evaluation techniques including confusion matrix precision recall ROC curves and cross validation
- KU5.** Importance of feature selection dimensionality reduction and data distribution in ML workflows

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Analyze and preprocess real world datasets for regression classification and clustering tasks

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- GS2.** Select appropriate ML algorithms based on problem type and data characteristics
- GS3.** Build train and fine tune models to optimize accuracy and generalization
- GS4.** Evaluate models using relevant performance metrics and validation techniques
- GS5.** Interpret results and present actionable insights from model outputs

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	7	7	-	6
PC1. Explain key machine learning concepts, types of learning, and common algorithms.	1	1	-	1
PC2. Build and evaluate regression models using techniques like linear and polynomial regression on real-world data.	1	1	-	1
PC3. Apply classification algorithms such as logistic regression, decision trees, and SVM to labeled datasets.	1	1	-	1
PC4. Use non-parametric techniques like K-Nearest Neighbors for classification tasks with unknown data distributions.	1	1	-	1
PC5. Perform clustering and dimensionality reduction using unsupervised learning methods like k-means and PCA.	1	1	-	1
PC6. Evaluate model performance using metrics such as confusion matrix, cross-validation, precision-recall, and ROC curves.	2	2	-	1
NOS Total	7	7	-	6

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National Occupational Standards (NOS) Parameters

NOS Code	ASP/ITS/N0105
NOS Name	Core Concepts in Machine Learning
Sector	Information Technology Sector
Sub-Sector	
Occupation	Artificial Intelligence & Machine Learning, Artificial Intelligence & Machine Learning
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

ASP/ITS/N0103: Bayesian methods And Text Mining

Description

Learners will develop the ability to implement Bayesian inference, probabilistic modeling, and text mining methods using Python for extracting patterns and solving uncertainty-based problems.

Scope

The scope covers the following :

- Learners will develop the ability to implement Bayesian inference, probabilistic modeling, and text mining methods using Python for extracting patterns and solving uncertainty-based problems.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Apply Bayesian inference concepts including prior, likelihood, and posterior in probabilistic models.
- PC2.** Implement Gaussian Mixture Models using Python to represent and analyze complex data distributions.
- PC3.** Use MCMC, Gibbs sampling, or variational inference to approximate posterior distributions in Bayesian models.
- PC4.** Preprocess textual data through tokenization, stop-word removal, and normalization techniques.
- PC5.** Extract features from text using TF-IDF, word embeddings, or other vectorization techniques.
- PC6.** Perform sentiment or topic analysis on unstructured text using appropriate Python libraries.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	7	7	-	6
PC1. Apply Bayesian inference concepts including prior, likelihood, and posterior in probabilistic models.	1	1	-	1
PC2. Implement Gaussian Mixture Models using Python to represent and analyze complex data distributions.	1	1	-	1
PC3. Use MCMC, Gibbs sampling, or variational inference to approximate posterior distributions in Bayesian models.	1	1	-	1
PC4. Preprocess textual data through tokenization, stop-word removal, and normalization techniques.	1	1	-	1
PC5. Extract features from text using TF-IDF, word embeddings, or other vectorization techniques.	1	1	-	1
PC6. Perform sentiment or topic analysis on unstructured text using appropriate Python libraries.	2	2	-	1
NOS Total	7	7	-	6

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National Occupational Standards (NOS) Parameters

NOS Code	ASP/ITS/N0103
NOS Name	Bayesian methods And Text Mining
Sector	Information Technology Sector
Sub-Sector	
Occupation	Artificial Intelligence & Machine Learning
NSQF Level	5
Credits	1
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

ASP/ITS/N0103: Python Programming With Scientific And Mathematical Applications

Description

Learners will acquire foundational to intermediate proficiency in Python, enabling them to apply programming skills to scientific and mathematical problem-solving, data processing, and model development using essential libraries.

Scope

The scope covers the following :

- The trainee shall be capable of doing Python programming using scientific Python libraries

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Write and debug Python programs using correct syntax data types control structures and file operations
- PC2.** Use NumPy and SciPy libraries to perform array manipulations and scientific computations
- PC3.** Create and interpret visualizations using Matplotlib for scientific data
- PC4.** Connect to databases using Python and perform data insertion or retrieval and updates
- PC5.** Solve linear algebra problems using Python for scientific and engineering applications
- PC6.** Simulate and analyze random processes using appropriate Python libraries and techniques
- PC7.** Formulate and solve basic optimization problems using tools like SciPy and PuLP

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understanding of Python syntax data types and control structures
- KU2.** Familiarity with NumPy and SciPy libraries for array manipulations and scientific computations
- KU3.** Knowledge of Matplotlib for creating and interpreting visualizations
- KU4.** Understanding of database connectivity using Python
- KU5.** Familiarity with linear algebra concepts and their application in Python

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Problem solving skills
- GS2.** Data analysis and interpretation skills
- GS3.** Programming skills in Python

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GS4. Data visualization skills

GS5. Critical thinking and analytical skills

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	7	7	-	6
PC1. Write and debug Python programs using correct syntax data types control structures and file operations	1	1	-	1
PC2. Use NumPy and SciPy libraries to perform array manipulations and scientific computations	1	1	-	1
PC3. Create and interpret visualizations using Matplotlib for scientific data	1	1	-	1
PC4. Connect to databases using Python and perform data insertion or retrieval and updates	1	1	-	1
PC5. Solve linear algebra problems using Python for scientific and engineering applications	1	1	-	1
PC6. Simulate and analyze random processes using appropriate Python libraries and techniques	1	1	-	1
PC7. Formulate and solve basic optimization problems using tools like SciPy and PuLP	1	1	-	-
NOS Total	7	7	-	6

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National Occupational Standards (NOS) Parameters

NOS Code	ASP/ITS/N0103
NOS Name	Python Programming With Scientific And Mathematical Applications
Sector	Information Technology Sector
Sub-Sector	
Occupation	Artificial Intelligence & Machine Learning, Artificial Intelligence & Machine Learning
NSQF Level	5
Credits	3
Version	1.0
Next Review Date	NA

Qualification Pack

ASP/ITS/N0106: Deep Learning Foundations to Prompt Engineering

Description

Learners will develop a practical and theoretical understanding of deep learning architectures and gain hands-on experience with neural networks, generative models, transformers, LLMs, and prompt engineering for advanced AI applications.

Scope

The scope covers the following :

- Covers deep learning basics and prompt engineering for AI solutions.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Explain the basic concepts and significance of deep learning and neural networks in modern AI
- PC2.** Describe the components and flow of a neural network including neurons layers and activation functions
- PC3.** Implement backpropagation algorithms for training neural networks on structured datasets
- PC4.** Apply regularization methods and dropout to prevent overfitting in deep learning models
- PC5.** Build and evaluate convolutional and recurrent neural networks for tasks such as image recognition and sequence prediction
- PC6.** Explain the principles and applications of generative models including VAEs GANs and GPTs

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** understand the core principles of deep learning neural networks and their role in artificial intelligence
- KU2.** know the structure and function of neurons layers activation functions and model architectures
- KU3.** understand training processes including backpropagation optimization algorithms and regularization techniques
- KU4.** know the applications and working of convolutional recurrent and transformer based models in various domains
- KU5.** understand prompt engineering methods and their importance in guiding large language models for specific tasks

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate technical concepts and model outcomes clearly in both written and verbal form

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- GS2.** work collaboratively in a team to design build and evaluate AI models
- GS3.** perform data analysis and calculations accurately when preparing datasets and evaluating models
- GS4.** use programming tools and deep learning frameworks effectively for model development
- GS5.** apply problem solving skills to debug model issues and refine prompts for improved AI performance

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	7	7	-	6
PC1. Explain the basic concepts and significance of deep learning and neural networks in modern AI	1	1	-	1
PC2. Describe the components and flow of a neural network including neurons layers and activation functions	1	1	-	1
PC3. Implement backpropagation algorithms for training neural networks on structured datasets	1	1	-	1
PC4. Apply regularization methods and dropout to prevent overfitting in deep learning models	1	1	-	1
PC5. Build and evaluate convolutional and recurrent neural networks for tasks such as image recognition and sequence prediction	1	1	-	1
PC6. Explain the principles and applications of generative models including VAEs GANs and GPTs	2	2	-	1
NOS Total	7	7	-	6

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National Occupational Standards (NOS) Parameters

NOS Code	ASP/ITS/N0106
NOS Name	Deep Learning Foundations to Prompt Engineering
Sector	Information Technology Sector
Sub-Sector	
Occupation	Artificial Intelligence & Machine Learning, Artificial Intelligence & Machine Learning
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

ASP/ITS/N0108: AI in the Cloud: Deployment, Ethics & Intelligent Agents.

Description

Learners will develop competency in deploying AI models on cloud platforms, ensuring ethical practices, and designing agent-based systems for real-world intelligent automation.

Scope

The scope covers the following :

- Learners will develop competency in deploying AI models on cloud platforms, ensuring ethical practices, and designing agent-based systems for real-world intelligent automation.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Explain core cloud computing concepts and their application in deploying AI/ML models at scale.
- PC2.** Utilize AI/ML services on AWS, Azure, or Google Cloud to train and deploy machine learning workflows.
- PC3.** Identify ethical risks in AI systems and apply practices to ensure fairness, transparency, and data privacy.
- PC4.** Use Hugging Face tools to fine-tune, deploy, and serve pre-trained NLP or generative models on the cloud.
- PC5.** Plan and execute an end-to-end capstone project using unsupervised or combined techniques.
- PC6.** Demonstrate the deployment of AI models using inference APIs and managed environments.
- PC7.** Implement basic autonomous agent systems that perform planning, tool usage, and task execution.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	7	7	-	6
PC1. Explain core cloud computing concepts and their application in deploying AI/ML models at scale.	1	1	-	1
PC2. Utilize AI/ML services on AWS, Azure, or Google Cloud to train and deploy machine learning workflows.	1	1	-	1
PC3. Identify ethical risks in AI systems and apply practices to ensure fairness, transparency, and data privacy.	1	1	-	1
PC4. Use Hugging Face tools to fine-tune, deploy, and serve pre-trained NLP or generative models on the cloud.	1	1	-	-
PC5. Plan and execute an end-to-end capstone project using unsupervised or combined techniques.	1	1	-	1
PC6. Demonstrate the deployment of AI models using inference APIs and managed environments.	1	1	-	1
PC7. Implement basic autonomous agent systems that perform planning, tool usage, and task execution.	1	1	-	1
NOS Total	7	7	-	6

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ASP/ITS/N0108
NOS Name	AI in the Cloud: Deployment, Ethics & Intelligent Agents.
Sector	Information Technology Sector
Sub-Sector	
Occupation	Artificial Intelligence & Machine Learning
NSQF Level	5
Credits	1
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

ASP/ITS/N0109: On the Job Training for AI ML Developer.

Description

Learners will acquire real-world experience by participating in on-the-job training aligned with AI workflows, providing practical exposure to tools, practices, and collaborative environments used in AI/ML deployments.

Scope

The scope covers the following :

- Learners will acquire real-world experience by participating in on-the-job training aligned with AI workflows, providing practical exposure to tools, practices, and collaborative environments used in AI/ML deployments.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Internship

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	-	170	10
PC1. Internship	-	-	170	10
NOS Total	-	-	170	10

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National Occupational Standards (NOS) Parameters

NOS Code	ASP/ITS/N0109
NOS Name	On the Job Training for AI ML Developer.
Sector	Information Technology Sector
Sub-Sector	
Occupation	Artificial Intelligence & Machine Learning
NSQF Level	5
Credits	6
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

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Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

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- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/09/2026
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Assessment Framework

ASAP follows a structured assessment system covering Attendance, Assignments, Internal Assessments, and Final Assessment. Attendance and internal assessments come under Continuous and Comprehensive Evaluation (CCE). All results are recorded as marks and reported as overall grades.

Weightage (NSQF Level 4 & above – Knowledge-based courses):

Practical: 60%

Theory: 40%

Qualification Pack

Assessment Components

Theory (Max 300) Marks Practical (Max 300) Marks

Assignment 1 40 Practical Assignment 40

Internal Assessment 1 60 Internal Practical Assessment 60

Assignment 2 40 Internship/OJT 100

Internal Assessment 2 60 Viva 90

Final Assessment 100 Attendance 10

1.2 Attendance Criteria

Daily attendance is mandatory due to the practical nature of skill sessions.

Eligibility: Students with below 70% attendance cannot appear for final assessment.

Attendance % Marks

Above 90 10

86-90 8

80-85 5

70-80 3

Below 70 0

2. Internal Assessment & Student Involvement

2.1 Continuous & Comprehensive Evaluation (CCE)

Includes Assignments, Internal Tests, and Attendance.

2.2 Assessment Plan

Logo
Not
Available



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Continuous Assessment:

Four knowledge/application tests

Attendance, Assignments, Internal Tests

Course-End Assessment:

Practical/skill test

Project & Viva

Final theory exam (MCQ + descriptive)

2.3 Assignments

Two assignments: after 25% and 75% syllabus completion

No restriction on reference materials

Scores entered twice in the student profile

2.4 Internal Assessments

Two tests: after 50% and 100% of syllabus

Question mix: 60% objective, 40% descriptive

Logo
Not
Available



Qualification Pack

Duration: 1 hour (mid-test) and 3 hours (final internal test)

Grades entered at both stages

3. Responsibility of Assessment Division

Final Assessment

Conducted by ASAP Kerala

Question paper generated randomly from the Question Bank

Evaluation done by external experts; results forwarded for certification

3.1 & 3.2 Question Bank

Developed by SMEs

Contains 6x the required questions

20% questions replaced annually

Blueprint used to ensure proper weightage and difficulty levels

3.3 Question Paper Generation

Batches receive unique login credentials

Qualification Pack

Trainers get access only at scheduled assessment time

Randomized generation as per competency requirements

3.4 Grading Scheme

Score Grade

90-100 A+

80-89 A

70-79 B

60-69 C

50-59 P

Below 50 F

Minimum Aggregate Passing % at QP Level : 50

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ASP/ITS/N0107.Python Programming with Scientific and Mathematical Applications	7	7	0	6	20	20
ASP/ITS/N0105.Core Concepts in Machine Learning	7	7	0	6	20	20
ASP/ITS/N0103.Bayesian methods And Text Mining	7	7	0	6	20	15

Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ASP/ITS/N0106.Deep Learning Foundations to Prompt Engineering	7	7	0	6	20	20
ASP/ITS/N0108.AI in the Cloud: Deployment, Ethics & Intelligent Agents.	7	7	0	6	20	20
ASP/ITS/N0109.On the Job Training for AI ML Developer.	0	0	170	10	180	5
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	0
Total	55	65	170	40	330	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
AA	Assessment Agency
AB	Awarding Body
ISCO	International Standard Classification of Occupations
NCO	National Classification of Occupations
NCrF	National Credit Framework
NOS	National Occupational Standard
NQR	National Qualification Register
NSQF	National Skill Qualification Framework
OJT	On the Job Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
National Occupational Standard	NOS define the measurable performance outcomes required from an individual engaged in a particular task. They list down what an individual performing that task should know and also do.
Qualification	A formal outcome of an assessment and validation process which is obtained when a competent body determines that an individual has achieved learning outcomes to given standards
Qualification File	A Qualification File is a template designed to capture necessary information of a Qualification from the perspective of NSQF compliance. The Qualification File will be normally submitted by the awarding body for the qualification.
Sector	A grouping of professional activities on the basis of their main economic function, product, service or technology.
Long Term Training	Long-term skilling means any vocational training program undertaken for a year and above.